Week 9: 26 – 30 October 2026

Subject: MYP Design (Year 2) **Unit Title:** Designing Ethical AI Systems – Developing Ideas **Grade: 7**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: **Generating Two AI Design Ideas**	• Sketch two different AI design ideas for a real school problem, each with at least 4 labels (input, process, output, user). • Compare both ideas against 3 design specifications written last week. • Score how realistic (feasible) each idea is and name one trade-off.	**Formative:** Two AI Design Ideas worksheet – teacher feedback against MYP Criterion B (strand ii). Not graded; it prepares students for the Blueprint summative. **Check for learning:** 3-2-1 exit ticket and self-rating against MUST / SHOULD / COULD. **Differentiation:** Support – picture cards, sentence starters and word bank; Challenge – explain a trade-off and how each idea could be biased.	Lesson 1 PowerPoint (W9.1) – on Toddle Two AI Design Ideas worksheet – on Toddle Video: "Crazy 8s" fast-sketching technique (YouTube)
Lesson 2 **Creating the AI Blueprint / Storyboard (Draft)**	• Choose the stronger idea and explain why, using at least 2 design specifications. • Draft a blueprint (flowchart) or storyboard showing at least 5 steps from input to output. • Label 3 data inputs and 2 outputs, and show 1 fairness or safety check.	**Formative:** Teacher observation, questioning, blueprint draft check and peer feedback (two stars and a wish). Not graded. **Check for learning:** 3-2-1 exit ticket and self-rating against MUST / SHOULD / COULD. **Differentiation:** Support – cut-out step cards and 4-box storyboard frame; Challenge – add a "human-in-the-loop" path and list the hardware, data and software needed.	Lesson 2 PowerPoint (W9.2) – on Toddle AI Blueprint worksheet – on Toddle Video: "A.I. Experiments: Teachable Machine" – Google (YouTube)
Homework	• **Lesson 1:** Upload photos of the two annotated ideas, comparison table and feasibility scorecard to Toddle. • **Lesson 2:** Act on the partner's "wish", then upload the draft blueprint/storyboard to Toddle.	Checked on Toddle by the teacher; feedback is given before the Blueprint summative.	Toddle – Week 9 assignments
Teacher's notes/ Reminders	• Students should bring coloured pencils, a ruler and a charged device to both lessons. • This week's work is practice (formative) and builds directly towards the Criterion B Blueprint summative. • Please ask your child: "What problem in our school does your AI solve, and how does it keep students' data safe?"		

Week 9

Technology Integration	Student devices, Toddle (work submission), Mentimeter (exit tickets), School AI, short YouTube clips, Lesson 1 & 2 PowerPoints.
Cross-curricular links	Computer Science (how AI uses data and algorithms); Ethics / Moral Education (fairness and privacy).
External Assessment Integration	Students practise MYP Criterion B (Developing Ideas) skills that are assessed in the IB MYP Design ePortfolio.
Wellbeing/ ATL Skills	• **Thinking:** creative idea generation; critical feasibility scoring, trade-offs and bias checks. • **Communication:** annotated sketches and blueprints others can understand. • **Collaboration / Wellbeing:** think–pair–share and kind, specific peer feedback (two stars and a wish).
Standards	• **MYP Design Criterion B:** strand ii – a range of feasible design ideas; strands iii–iv – chosen design justified and planning drawings. • **CS.MS.5.2:** The behaviour of AI systems reflects both the goals of the designers and the data used to train the system. • **CS.MS.1.1:** Complex problems can be broken into smaller parts to facilitate program implementation and review.
Inquiry/ TOK Question	**Factual:** What are the inputs, processes and outputs of an AI system? **Conceptual:** How do design specifications help designers choose the best idea? **Debatable:** Should the most accurate AI idea always be chosen, even if it is less fair?
Success Criteria (Measurable)	The student has met the learning targets by: • Producing two distinct, annotated AI design ideas (Lesson 1). • Comparing the ideas against their design specifications and identifying a trade-off (Lesson 1). • Showing the steps of their AI system with key inputs and outputs labelled (Lesson 2). • Showing where a fairness or safety check happens in the process (Lesson 2).
National Identity Domain	Culture – Promotion of innovation and responsible technology that serves the school community, in line with the UAE's vision for a secure, knowledge-based economy. Values – Respect, fairness and protecting others' privacy: students design AI that avoids bias and keeps people's data safe.